

Project: Task Manager

Problem Statement

Develop a Python-based Task Manager that allows users to organize their daily tasks. The program should enable users to add tasks, view all tasks, mark tasks as completed, and manage task information using appropriate Python data structures and functions.

Objective

Build a simple task management application to practice Python functions, lists, dictionaries, loops, conditional statements, and modular programming.

Requirements

- Create an empty task list.
- Each task should contain an ID, description, and completion status.
- Create a function to add a new task.
- Create a function to display all tasks.
- Create a function to search for a task by ID.
- Create a function to mark a task as completed.
- Display meaningful messages for invalid task IDs.
- Demonstrate the program by adding multiple tasks and updating their status.

Concepts Covered

- Functions
- Lists
- Dictionaries
- Loops
- Conditional Statements
- User-defined Functions
- Basic CRUD Operations

Expected Learning Outcomes

After completing this project, students will be able to create reusable functions, manage data using lists and dictionaries, perform CRUD operations, and build a simple console-based application in Python.